

# Customer Churn Analytics — Business Report

Prepared for: Telecom Executive Leadership

## Executive Summary

The telecom business faces a 26.54% churn rate, with 1,869 churned customers representing an estimated \$139,131 in monthly revenue loss. Contract structure, tenure, internet service type, and payment method are the strongest churn drivers. Logistic Regression achieved 84.25% ROC-AUC for proactive retention targeting.

## Key Metrics

| Metric                   | Value     |
|--------------------------|-----------|
| Total customers          | 7,043     |
| Churn rate               | 26.54%    |
| Monthly revenue (active) | \$316,986 |
| Revenue lost to churn    | \$139,131 |
| Average CLV              | \$2,280   |

## Model Comparison

| Model               | Accuracy | Precision | Recall | ROC-AUC |
|---------------------|----------|-----------|--------|---------|
| Logistic Regression | 79.91%   | 65.22%    | 52.14% | 84.25%  |
| XGBoost             | 79.13%   | 63.42%    | 50.53% | 83.68%  |
| Random Forest       | 78.28%   | 61.56%    | 48.40% | 82.50%  |

## Business Recommendations

- Target high-value customers with churn probability above 0.7 for retention offers.
- Promote annual/two-year contracts for month-to-month subscribers after 6 months.
- Migrate electronic check users to autopay or card billing.
- Launch a 90-day onboarding program for customers in their first year.
- Review fiber pricing and bundle support/streaming perks to reduce churn.